

Progress Report

Depth of Field Research

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Progress

- Built an off-screen framebuffer path: the scene renders into color and depth textures, then a full-screen post-processing pass.
- Sampled and linearized the depth texture inside the post-processing shader.
- Implemented a physical thin-lens circle of confusion (focus distance, focal length, f-number, sensor size; 1 unit = 1 m), with magnitude and signed near/far visualizations.
- Produced the first working DoF effect: a disk-gather blur whose radius comes from each pixel's CoC, keeping in-focus pixels sharp.
- Observed the limits of this simple approach: halos and color bleeding at depth edges between foreground and background.

Next Steps

- Research traditional DoF techniques and the current state of the art.
- Build a richer 3D scene to test DoF methods on.